

Nathan Gomes

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EDUCATION

University of Maryland

Bachelor of Science in Information Science, GPA: 3.96

College Park, MD

August 2023 – May 2027

EXPERIENCE

Hack4Impact

October 2024 – December 2024

University of Maryland (10 hours/week)

College Park, MD

- Designed and deployed full-stack web applications using Python, Node.js, and PostgreSQL for nonprofit organizations, serving 1,000+ community members.
- Collaborated in cross-functional agile teams, leveraging Git, code reviews, and sprint workflows to ship production-ready features on schedule.
- Performed database validation, application testing, debugging, and troubleshooting to improve system performance and reduce technical issues.

Data Collection Temp

January 2024 – March 2024

Leidos (18 hours/week)

McLean, VA

- Captured, structured, and validated large-scale roadway test data to support the development of machine-learning models for real-time intersection hazard detection in the Intersection Safety Challenge (ISC).
- Troubleshoot and optimized Linux-based data acquisition systems, improving data reliability and minimizing collection downtime during field testing.
- Assisted in capturing and organizing large-scale roadway test data, supporting the development of machine-learning models for real-time intersection hazard detection.

PROJECTS

Active Directory Lab | *Windows Server 2019, PowerShell, Oracle VirtualBox*

May 2026 – June 2026

- Built an Active Directory lab in Oracle VirtualBox using Windows Server 2019 and Windows client computers.
- Created and managed user accounts, groups, organizational units (OUs), and permissions.
- Used PowerShell scripts to automate the creation and management of 1,000+ Active Directory user accounts.
- Configured DNS, DHCP, and Active Directory Domain Services (AD DS) to support domain authentication.
- Troubleshoot user login, network connectivity, and domain-related issues.

Transportation Data Analytics Project | *Python, SQL, Pandas*

March 2026 – May 2026

- Developed a Python and SQL analytics pipeline to process 100,000+ Uber ride records, identifying demand trends, peak travel times, and high-traffic locations.
- Cleaned, transformed, and visualized large datasets to generate actionable insights through exploratory analysis and interactive reporting.

Traffic Citation Analytics Database | *SQL*

August 2025 – December 2025

- Led a 4-member team in designing and implementing a fully normalized relational database (11 tables, 3NF) to manage Chicago traffic citation data.
- Developed complex SQL queries, views, stored procedures, and aggregate functions to analyze citation trends by ZIP code, violation type, and issue date.
- Generated trend reports identifying high-frequency violation areas and average fines, supporting data-driven insights.
- Created ER diagrams and documentation to ensure scalability, data integrity, and efficient schema design.

TECHNICAL SKILLS & TECHNOLOGIES

Programming Languages: Python, SQL, R, JavaScript, HTML/CSS

Data Analysis & Visualization: Tableau, Microsoft Excel, Pandas

Tools & Technologies: Active Directory, Windows Server 2019, PowerShell, Git, GitHub, VS Code, RStudio, Oracle VirtualBox, Linux, Wireshark, Cisco Packet Tracer

IT Support: PC Troubleshooting, Hardware/Software Installation, System Configuration, User Account Management